HAT-P-32b

R-0188

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-32_241020` · created 2026-07-16T20:29:46+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460603.8712 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.007
Transit depth (Rp/Rs)^2	2.19 +/- 0.21 [%]
Orbital Inclination (inc)	89.94 +/- 1.01
Airmass coefficient 1 (a1)	1.2105 +/- 0.0091
Airmass coefficient 2 (a2)	-0.1444 +/- 0.0098
Scatter in the residuals of the lightcurve fit is	0.75 %
Best Comparison Star	None
Optimal Aperture	3.96
Optimal Annulus	9.72
Transit Duration (day)	0.1298 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.148 ± 0.007 vs 0.1489 (deviation 0.08σ)
Duration fit vs published	0.1298 d vs 0.1304 d (deviation 0.36σ)
Mid-time O–C	1.1 min
Depth SNR	13.5
Residual scatter	0.75 %

Light curve

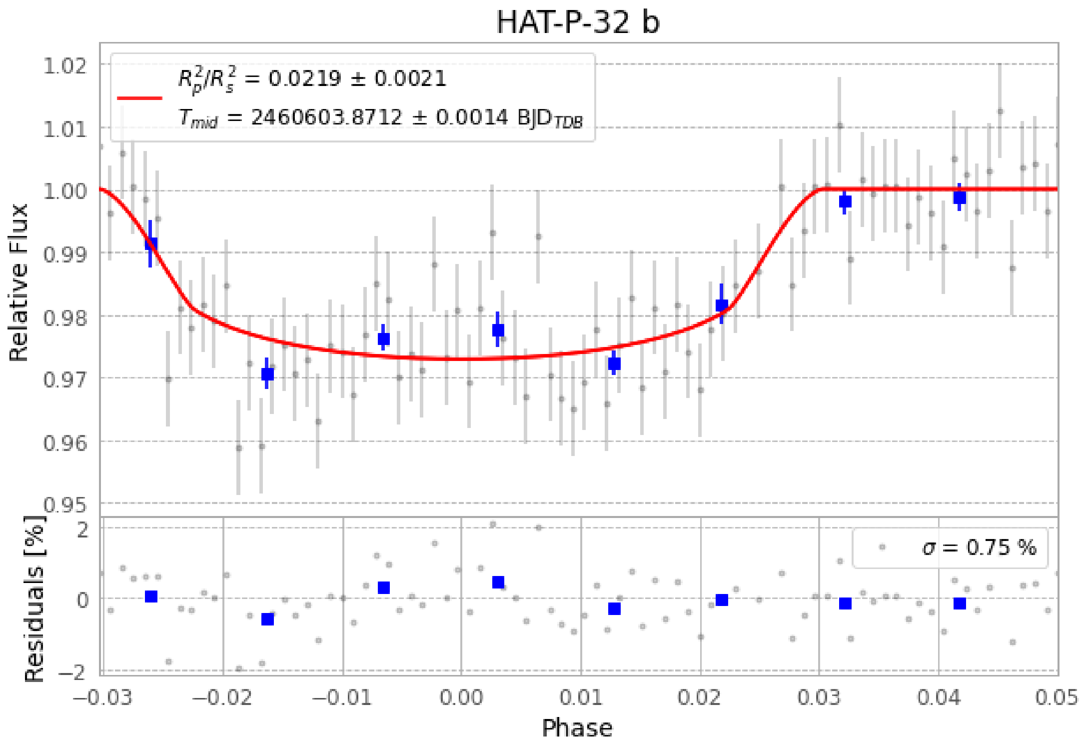

Figure 1 — FinalLightCurve_HAT-P-32 b_2024-10-20.png (SHA-256 d3ef2643acab6902…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [395, 265], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 4cacfecff695d51e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 378b18d`

Data provenance

84 FITS frames (55 MB), `HATP-32241020071215.FITS` → `HATP-32241020112115.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-241020.log (62829232b47c…), FinalLightCurve_HAT-P-32 b_2024-10-20.png (d3ef2643acab…), FinalLightCurve_HAT-P-32 b_2024-10-20.csv (00bbe7d2cf99…)

Compute resources

Wall time 185.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)